# Thymosin Beta 4 Available for Research Use Only

## (2-2.5 mg 2×/wk for 4–6 wk, then 2-2.5 mg 1×/wk until healed)

Full-length thymosin beta 4 (Tβ4) is a naturally occurring 43-amino-acid peptide involved in actin regulation and tissue repair. Human research has evaluated synthetic Tβ4 in topical wound and eye treatments and in intravenous safety studies. Animal evidence makes dermal repair, cardiac protection, neurological recovery, and hair-follicle support likely. Human therapeutic findings are preliminary and mixed; Broader injury-recovery effects remain plausible. [1,3,5-17]

💉 💉 **Probably safe:** Synthetic Tβ4 was well tolerated during a small, short-term intravenous studies in healthy volunteers. This supports short-term safety, rather than a general conclusion about long-term use. [3]

⭐ ⭐ **Probably effective** for dermal wound repair and tendon healing based on animal evidence. Broader tissue-repair benefits are plausible.

**Cancer caution:** Tβ4 participates in angiogenesis and cell migration, and some tumor models show increased metastatic activity. A potential effect on an existing malignancy warrants caution, although findings differ by cancer type. [20,21]

*Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.*

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Mature native Tβ4 is an N-terminally acetylated peptide of approximately 4.9 kDa. Its actin-binding surface extends across much of the molecule, allowing it to sequester globular actin and participate in controlled actin exchange. The full sequence also contains the N-terminal Ac-SDKP precursor region. These properties distinguish it from isolated short peptide fragments. [1,2,22]

### Gray Market Confusion Between Tβ4 and TB-500

The names **Tβ4** and **TB-500** are used inconsistently in gray market and research-use-only sales. In analytical research, TB-500 has been identified as **Ac-LKKTETQ**, a synthetic N-acetylated seven-amino-acid peptide corresponding to residues 17–23 of full-length thymosin beta 4. Full-length Tβ4 contains 43 amino acids. These are distinct molecules, and findings for one should not automatically be attributed to the other.

Vendor labels may use “TB-500” for either the fragment or full-length Tβ4. Labels such as “TB-500 (TB4)” or “TB-500 (frag)” can help indicate the intended identity, but the name alone does not

establish what a sample contains. **Ac-SDKP**, the acetylated four-amino-acid peptide derived from Tβ4's N-terminus, is another distinct molecule.

Janoshik's reporting terminology is one source which helps clarify these distinctions, and some vendors have adopted their language.

| Janoshik Label | Identity |
|---|---|
| **TB-500 (TB-4)** | Full-length, 43-amino-acid Tβ4 |
| **TB-500 889 (Da)** | The acetylated fragment corresponding to residues 17–23, Ac-LKKTETQ |
| **TB-500** | A fragment; consult the report's comments for its specific identity |

Verify peptide identity using a COA. Check the reported identity and comments, rather than relying on the product name or purity percentage alone. A product labeled "TB-500" may contain full-length Tβ4 or a fragment. Identify the molecule before using it.

## Dosing Patterns

Ruff and colleagues studied 42, 140, 420, and 1260 mg in four cohorts totaling 40 healthy volunteers. Participants first received a single dose of active treatment or placebo; after safety review, the assigned regimen was repeated daily for 14 days. These unusually large milligram doses are as reported in the publication. They describe phase I safety testing, not a loading or maintenance protocol for injury recovery. [3]

No evidence-based human dose or dosing interval has been established for subcutaneous Tβ4. Community protocols commonly use a **higher-dose initial loading phase** followed by **less intense maintenance dosing**. The following patterns are speculative community practices:

- **Mild injury and general recovery**: Loading, 2 mg twice weekly for 4 weeks. Maintenance, 2 mg once weekly during continued recovery.
- **Musculoskeletal injury**: Loading, 2.5 mg twice weekly for 4 to 6 weeks. Maintenance, 2 to 2.5 mg once weekly during continued recovery.
- **Severe injury or cardiac recovery**: Loading, 3 mg twice weekly for 6 weeks. Maintenance, 2.5 mg once weekly during continued recovery.

Sometimes twice-weekly dosing is divided into a similar weekly amount as daily doses.


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## Common Stacking

Combining full-length Tβ4 with BPC-157 is an experimental, albeit common, repair strategy. Synergies in the combination are plausible, but the combination requires direct evaluation of safety and additional benefit.

## Benefits

Tβ4 research developed from its role in the actin cytoskeleton into studies of wound repair, vascular growth, and cell survival. Full-length peptide experiments subsequently investigated cardiac injury, corneal damage, neurological injury, and hair growth. The resulting clinical programs include synthetic Tβ4 ophthalmic solution RGN-259, topical wound treatment, and intravenous formulations. [3,5,6,9-17]

The clinical record contains encouraging local-treatment signals, but also some unsuccessful primary endpoints. Small positive studies and secondary analyses support further plausible repair benefits, but they should be read together with some negative findings. [5-8,26-28]

**Dermal wound repair:** Tβ4 is likely to accelerate reepithelialization and wound contraction based on animal studies. A topical human venous-ulcer trial also reported a preliminary healing signal, particularly for smaller or less severe wounds. This supports local wound-healing research without establishing systemic treatment for all injuries. [5,9]

**Corneal surface repair and ocular comfort: T**opical Tβ4 is likely to support corneal repair based on animal injury studies. Human trials show selected improvements in ocular signs, comfort, or defect healing, but efficacy is mixed. The 72-patient dry-eye trial missed both primary endpoints while improving selected secondary endpoints; a separate nine-patient study reported improvements at selected visits. [6,7,13]

In SEER-1, complete healing at four weeks occurred in 6 of 10 treated patients and 1 of 8 placebo patients. The larger SEER-3 trial missed its four-week primary healing endpoint. These results make ocular benefit promising but insufficiently consistent for an unqualified efficacy claim. [8,27,28]

**Tendon and ligament healing:** Benefits are plausible through cell migration, vascular support, and tissue remodeling. The skin and cardiac literature supplies biological support for repair, but the tendon-fragment experiments from the original survey do not establish a tendon-healing effect. [2,9-11]

**Skeletal muscle recovery:** Tβ4 plausibly supports recruitment of repair cells. Muscle-injury studies show endogenous Tβ4 activity and myoblast chemotaxis, but intravenous treatment in a separate mouse injury experiment did not enlarge regenerating muscle fibers at the measured time points. A broad claim of faster or stronger muscle recovery is speculative. [23,24]


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**Cardiac protection and repair:** Tβ4 is likely to support cell survival and repair after experimental myocardial injury, based on mouse studies and vascular mechanisms. A 2025 randomized recombinant-peptide trial in 96 STEMI patients found no significant overall infarct-size difference, with a favorable signal in the subgroup first treated within eight hours after PCI. Human cardiac benefit remains plausible. [10-12,26]

**Inflammatory signaling:** Tβ4 is likely to reduce selected inflammatory responses in injured tissues, supported by corneal, hepatic, and neurological animal studies. Its effect is tied to the model and tissue response rather than a universal suppression of immunity. [13,14,18]

**Neurological tissue repair:** Tβ4 is likely to improve functional recovery and support neurorestorative processes based on rat traumatic-brain-injury and stroke studies. Mouse demyelination studies also support oligodendrocyte maturation and remyelination. These are direct preclinical findings; human neurological recovery remains a plausible therapeutic application. [14-16]

**Hair follicle support:** Tβ4 is likely to stimulate hair growth through follicular stem-cell migration and differentiation based on rodent experiments. Benefits for human scalp-hair loss are plausible. [17].

## Potential Applications

**Acute soft tissue repair:** Dermal wounds are the most directly supported soft-tissue application, with both animal findings and early topical human data. Tendon, ligament, and postsurgical systemic repair remain plausible extensions. [5,9]

**Chronic overuse injuries:** Repair-cell migration and extracellular-matrix remodeling make benefits for chronic tendinopathy plausible. The rationale does not establish a treatment course or a reliable recovery timeline. [2,9]

**Ocular surface disorders:** Dry eye and neurotrophic keratopathy are active clinical-development applications for local Tβ4. Mixed primary outcomes support continued investigation with clearly defined populations and endpoints. [6-8,27,28]

**Post infarction cardiac repair:** Animal protection and the early-treatment signal with recombinant Tβ4 support continued study as an adjunct after reperfusion. This application remains investigational. [10,26]

**Neurological recovery:** Traumatic brain injury, ischemic stroke, and demyelinating disorders have direct animal support for functional or tissue recovery. Clinical neurological benefits are plausible. [14-16]


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**Hepatic injury and fibrosis:** Exogenous Tβ4 reduced injury and fibrosis in one rodent toxic-injury study. Conversely, selectively deleting endogenous Tβ4 from activated hepatic stellate cells reduced fibrosis in other mouse models. Administered peptide, cellular expression, and disease context can therefore have different effects. Hepatic antifibrotic treatment is plausible in selected settings, rather than uniformly likely. [18,19]

**Renal tissue repair:** A mouse obstructive-kidney model showed reduced late fibrosis with Tβ4, while peptide processing and timing influenced the response. Renal repair is a plausible human application. [25]

## Cautions

**Active or previous malignancy:** A plausible concern about supporting an existing tumor follows from Tβ4-related vascular growth and cell movement and from tumor-promoting experimental findings. Active cancer warrants particular caution and oncology involvement. Previous cancer requires assessment of current disease and recurrence risk. [20,21]

**Pregnancy and breastfeeding:** The studies summarized here do not establish reproductive, fetal, or lactation safety. Avoid experimental use during pregnancy or breastfeeding as a precaution.

**Children and adolescents:** The adult clinical regimens do not establish pediatric dosing or safety. Use in children or adolescents requires qualified clinical or research oversight.

## Side Effects

In the small synthetic Tβ4 intravenous safety trial, adverse events were infrequent and mild or moderate, with no serious adverse events or dose-limiting toxicity. The topical and ocular studies provide additional short-term tolerability observations for those local preparations. [3,5-8]

## Potential for Diminishing Benefit

Tβ4's intracellular actin-sequestering function differs from a conventional receptor-agonist mechanism. Clinical studies do not establish that cycling prevents diminished response or that maintenance treatment improves completed healing. [2,9-11]

## Biological Mechanism

**Actin regulation:** Full-length Tβ4 binds globular actin in a complex that prevents immediate filament incorporation. Exchange with profilin helps make actin available for controlled polymerization. These interactions support cytoskeletal changes involved in cell shape and movement. The molecule's extended binding interface is essential to understanding the full-length peptide. [2]


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**Cell migration and vascular repair:** Tβ4 influences migration of repair-associated cells and vascular progenitors. Full-length experiments link it to wound reepithelialization, collagen deposition, cardiac cell survival, and epicardial vascular responses. Improved blood supply during tissue repair is a plausible consequence of these vascular effects. [9-11]

**Cell survival and inflammatory regulation:** Preclinical studies implicate integrin-linked kinase and Akt-associated survival signaling in cardiac repair. Neurological and corneal models show tissue protection and modulation of injury-associated inflammation. These effects represent several interacting repair pathways, rather than an actin mechanism that explains every therapeutic finding. [10,13-16]

**N terminal peptide processing:** The acetylated N-terminal residues of native Tβ4 contain the Ac-SDKP sequence. Laboratory and animal experiments demonstrate release through sequential processing involving meprin-α and prolyl oligopeptidase. Ac-SDKP formation provides a plausible contribution to some anti-inflammatory and antifibrotic effects, but direct fragment administration and full-length administration have different exposures. [22,25]

## TB-500 and Cancer Risk

The cancer concern involves both repair biology and direct tumor-model findings. In a mouse melanoma experiment, increasing tumor-cell Tβ4 expression increased migration, tumor vascularization, and metastatic potential. This supports a context-specific tumor-promotion concern, but gene overexpression is not equivalent to administering an external peptide. [20]

The direction is not uniform. In multiple myeloma, increased Tβ4 expression reduced proliferation and migration in experimental cells and was associated with a protective pattern in mouse and patient-expression analyses. These contrasting results mean that neither universal cancer promotion nor universal protection is justified. [21]

Short clinical tolerability studies cannot resolve long-latency cancer risk. The concern is most relevant to an existing or recurrent malignancy and should be assessed in its disease context. A claim that Tβ4 initiates cancer in humans is not established by these findings. [3,4,20,21]

## Conclusions

Tβ4 is widely available as a for-research-use-only chemical, often labeled as TB-500. Full-length Tβ4 has a substantive research base in actin regulation and tissue repair. Dermal healing, cardiac protection, neurological recovery, and hair-follicle activity are likely based on direct animal evidence. Human studies provide short-term tolerability data and selected local-treatment signals, but missed primary endpoints. [2,3,5-17,26-28]

Tendon, ligament, general musculoskeletal, and systemic recovery applications remain plausible. Liver-fibrosis and cancer effects require attention to conflicting, context-specific


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findings. Community loading and maintenance doses and patterns should be read as unvalidated examples. [18-21,23,24]

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